KENDRIYA VIDYALAYA SANGTHAN, CHANDIGARH REGION
PREBOARD EXAM - 2024-25
CLASS: XII
SUBJECT: PHYSICS (THEORY)

Maximum Marks: 70 Time Allowed: 3 hours.

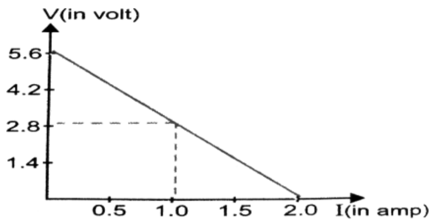
General Instructions:

(1) There are 33 questions in all. All questions are compulsory.
(2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
(3) All the sections are compulsory.
(4) Section A contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study based questions of four marks each and Section E contains three long answer questions of five marks each.
(5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
(6) Use of calculators is not allowed.
(7) You may use the following values of physical constants where ever necessary

i. $c = 3 \times 10^8$ m/s
ii. $m_e = 9.1 \times 10^{-31}$ kg
iii. $e = 1.6 \times 10^{-19}$ C
iv. $\mu_0 = 4\pi \times 10^{-7}$ TmA^{-1}
v. $h = 6.63 \times 10^{-34}$ Js
vi. $\varepsilon_0 = 8.854 \times 10^{-12}$ $C^2N^{-1}m^{-2}$
vii. Avogadro's number = 6.023×10^{23} per gram mole

SECTION A

1. A particle of mass m and charge q is placed at rest in uniform electric field E and then released, the kinetic energy attained by the particle after moving a distance y, will be
a) q^2Ey b) qEy c) qE^2y d) qEy^2
2. Which of the following is NOT the property of equipotential surface?
a) They do not cross each other.
b) The rate of change of potential with distance on them is zero.
c) For a uniform electric field they are concentric spheres.
d) They can be imaginary spheres.
3. A straight line plot showing the terminal potential difference (V) of a cell as a function of current (I) drawn from it is shown in figure.



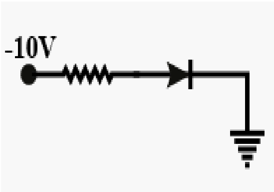
The internal resistance of the cell would be then
a) 2.8 ohm b) 1.4 ohm c) 1.2 ohm d) zero
4. An electric current passes through a long straight wire. At a distance of 5 cm from the wire, the magnetic field is B. The magnetic field at 20 cm from the wire would be
a) B/6 b) B/4 c) B/3 d) B/2

5. Among the following properties describing diamagnetism, identify the property that is wrongly stated.
   a) Diamagnetic materials do not have permanent magnetic moment.
   b) Diamagnetism is explained in terms of electro-magnetic induction.
   c) Diamagnetic materials have a small positive susceptibility.
   d) The magnetic moment of individual electrons neutralize each other.
6. If wattless current flows in the AC circuit, then the circuit is
   a) purely resistive circuit
   b) LCR series circuit
   c) RC series circuit
   d) purely inductive circuit.
7. Match list I with list II

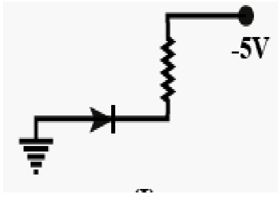
List I	List II
(p) UV rays (q) X- rays (r) Microwave (s) infrared wave	(i) diagnostic tool in medicine (ii) water purification (iii) communication, radar (iv) improving visibility in foggy days

   Choose the correct answer from the options given below:
   a) (p) – (iii), (q) – (ii), (r) – (i), (s) – (iv)
   b) (p) – (ii), (q) – (i), (r) – (iii), (s) – (iv)
   c) (p) – (ii), (q) – (iv), (r) – (iii), (s) – (i)
   d) (p) – (iii), (q) – (i), (r) – (ii), (s) – (iv)
8. A ray of laser of a wavelength 630nm is incident at an angle of 30° at the diamond-air interface. It is going from diamond to air. The refractive index of diamond is 2.42 and that of air is 1. Choose the correct option.
   a) angle of refraction is 24.41°
   b) angle of refraction is 30°
   c) refraction is not possible
   d) angle of refraction is 53.4°
9. If in Young's double slit experiment of light, interference is performed in water, which one of the following is correct?
   a) fringe width will decrease
   b) fringe width will increase
   c) there will be no fringe
   d) fringe width will remain unchanged.
10. A light having wavelength 300nm falls on a metal surface. Work function of metal is 2.54 eV. What is stopping potential.
   a) 2.3 V   b) 2.59 V   c) 1.60 V   d) 1.29 V
11. According to Bohr atom model, in which of the following transitions, will the frequency be maximum?
   a) n = 3 to n = 2
   b) n = 5 to n = 4
   c) n = 4 to n = 3
   d) n = 2 to n = 1

12. In the following figures, which one of the diodes is reverse biased?



a



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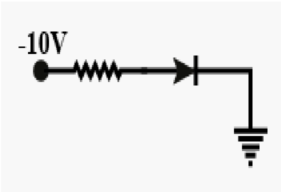

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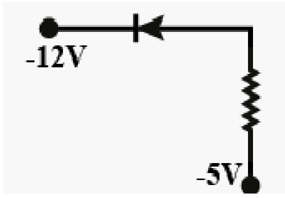


d

For Questions 13 to 16, two statements are given –one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

a) If both Assertion and Reason are true and Reason is correct explanation of Assertion.

b) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

c) If Assertion is true but Reason is false.

d) If both Assertion and Reason are false.

13. Assertion: To increase the range of an ammeter, we must connect a suitable high resistance in series with it.
Reason: The ammeter with increased range should have high resistance.

14. Assertion: To observe diffraction of light the size of obstacle/aperture should be of the order of 10^{-7}m.
Reason: 10^{-7}m is the order of wavelength of visible light.

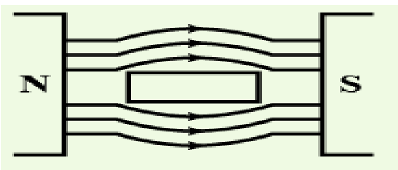
15. Assertion: For the radiation of a frequency greater than the threshold frequency, photo- electric current is proportional to the intensity of the radiation.
Reason: Greater the number of energy quanta available, greater is the number of electrons absorbing the energy quanta and greater is number of electrons coming out of the metal.

16. Assertion: Nuclei having mass number about 60 are most stable.
Reason: When two or more light nuclei are combined into a heavier nucleus, then the binding energy per nucleon will increase.

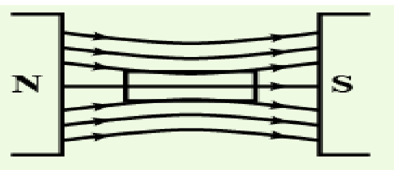
SECTION B

17. a) Obtained the conditions under which an electron does not suffer any deflection while passing through a magnetic field.
b) Two protons P and Q moving with same speed pass through the magnetic field $\mathbf{B}_1$ and $\mathbf{B}_2$ respectively, at right angles to field directions. If $B_2 > B_1$, which of the two protons will describe circular path of smaller radius? Explain.

18. A uniform magnetic field gets modified as shown in figure, when two specimens X and Y are placed in it.
(i) Identify the two specimens X and Y.
(ii) State the reason for the behaviour of the field lines in X and Y.



X



Y

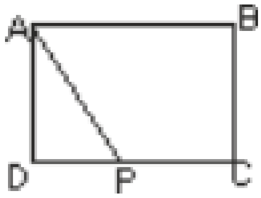
Or

The susceptibility of a magnetic material is 2.6×10^{-5}. Identify the type of magnetic material and state its two properties.

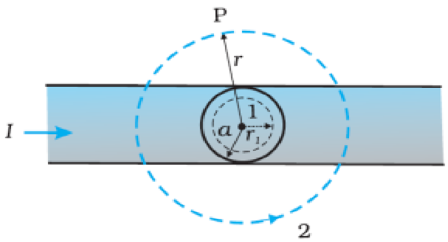
19. Identify the electromagnetic wave whose wavelength lies in the range
(a) $10^{-11}\text{m} < \lambda < 10^{-14}\text{m}$ and (b) $10^{-4}\text{m} < \lambda < 10^{-6}\text{m}$. Write any one use of each.
20. A hydrogen atom is in its third excited state.
(a) How many spectral lines can be emitted by it before coming to ground state? Show these transitions in energy level diagram.
(b) In which of the above transitions will the spectral line of shortest wavelength be emitted?
21. (a) Name the device which utilizes unilateral action of a pn diode to convert ac into dc. (b) Draw the circuit diagram of full wave rectifier.

SECTION C

22. A wire of uniform cross-section and resistance 4 ohm is bent in the shape of square ABCD. Point A is connected to a point P on DC by a wire AP of resistance 1 ohm. When a potential difference is applied between A and C, the points B and P are seen to be at the same potential. What is the resistance of the part DP?



23. The given figure shows a long straight wire of a circular cross-section (radius a) carrying steady current I. The current I is uniformly distributed across this cross-section. Calculate the magnetic field in the region r < a and r > a.

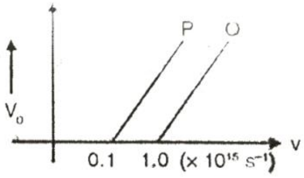


Or

Describe the working principle of moving coil galvanometer? Why it is necessary to use (i) a radial magnetic field and (ii) cylindrical soft iron core in a galvanometer?
Can a galvanometer as such be used for measuring the current?

24. (a) Define mutual inductance and write its SI unit.
(b) Two circular loops, one of small radius r and other of larger radius R, such that R >> r, are placed coaxially with centres coinciding. Obtain the mutual inductance of the arrangement.
25. An inductor, a capacitor and a resistor are connected in series with an alternating source $v = v_m \sin\omega t$. Derive an expression for the average power dissipated in the circuit. Also obtain the expression for the resonant frequency of the circuit.
26. Draw the ray diagram explaining the image formation in astronomical refracting type telescope. Write expression for magnification in normal adjustment. Write any two advantages of reflecting type telescope over refracting type telescope.
27. An electron and a Photon each have a wavelength 2 nm. Find
a) their momentum
b) the energy of photon and
c) the kinetic energy of electron.

28. Figure shows the variation of stopping potential V_0 with the frequency υ of the incidence radiation for two photosensitive metals P and Q.

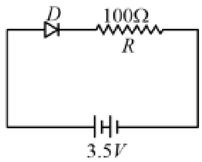


(i) Explain which metal has smaller threshold wavelength.
(ii) Explain giving reason which metal emits photoelectrons having smaller kinetic energy for the same wavelength of incident radiation.
(iii) If the distance between the light source and metal P is doubled, how will the stopping potential change?

SECTION D

29. Read the following paragraph and answer the questions that follow.
A semiconductor diode is basically a p-n junction with metallic contacts provided at the ends for the application of an external voltage. It is a two terminal device. When an external voltage is applied across a semiconductor diode such that p-side is connected to the positive terminal of the battery and n-side to the negative terminal, it is said to be forward biased. When an external voltage is applied across the diode such that n-side is positive and p-side is negative, it is said to be reverse biased. An ideal diode is one whose resistance in forward biasing is zero and the resistance is infinite in reverse biasing. When the diode is forward biased, it is found that beyond forward voltage called knee voltage, the conductivity is very high. When the biasing voltage is more than the knee voltage the potential barrier is overcome and the current increases rapidly with increase in forward voltage. When the diode is reverse biased, the reverse bias voltage produces a very small current about a few micro-amperes which almost remains constant with bias. This small current is reverse saturation current.

(i) In the given figure, a diode D is connected to an external resistance R = 100 Ω and an emf of 3.5 V. If the barrier potential developed across the diode is 0.5 V, the current in the circuit will be:



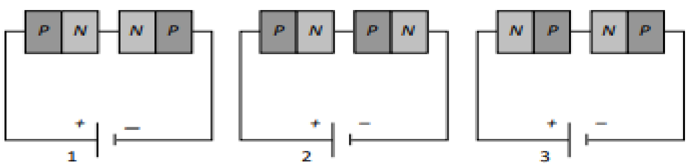
(a) 40 mA (b) 20 mA (c) 35 mA (d) 30 mA

(ii) In an ideal diode the resistance during forward bias will be
(a) infinite (b) zero (c) small (d) large

(iii) Based on the V-I characteristics of the diode, we can classify diode as
(a) bilateral device (b) ohmic device
(c) non-ohmic device (d) passive element

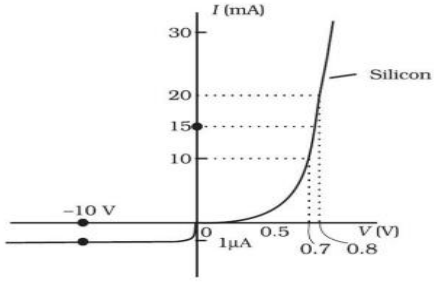
OR

Two identical PN junctions can be connected in series by three different methods as shown in the figure. If the potential difference in the junctions is the same, then the correct connections will be



(a) in the circuits (1) and (2) (b) in the circuits (2) and (3)
(c) in the circuits (1) and (3) (d) only in the circuit (1)

(iv)



The V-I characteristic of a diode is shown in the figure. The ratio of the resistance of the diode at I = 15 mA to the resistance at V = -10 V is

(a) 100 (b) 10^6 (c) 10 (d) 10^{-6}

30. transparent media of refractive indices n_1 and n_2 are separated by a spherical transparent surface. The rays of light incident on the surface get refracted into the medium on the other side. The laws of refraction are valid at each point of the spherical surface. A lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. The focal length of a lens is determined by the radii of curvature (R_1 and R_2) of its two surfaces and the refractive index (n) of the medium of the lens with respect to the surrounding medium. Depending on R_1 and R_2 a lens behaves as a diverging or a converging lens. The ability of a lens to diverge or converge a beam of light incident on it defines it power.

(i) Ac point object is placed in air at a distance R in front of a convex spherical refracting surface of radius of curvature R. If the medium on the other side of the surface is glass, then the image is

(a) real and formed in the glass.
(b) Real and formed in air
(c) Virtual and formed in glass
(d) Virtual and firmed in air

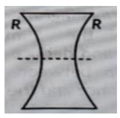
(ii) An object is kept at 2F in front of an equiconvex lens. The image formed is

a) real and of the size of the object
b) virtual and of the size of object
c) real and enlarged
d) virtual and diminished.

(iii) A thin converging lens of focal length 10cm and a thin diverging lens of focal length 20cm are placed coaxially in contact. The power of the combination is

(a) -5D (b) +5D (c) +15D (d) -15D

(iv) An equiconcave lens of focal length 'f' is cut into two identical parts along the dotted line as shown in the figure. The focal length of each part will be



(a) f/a (b) f/2 (c) f (d) 2f

OR

A spherical air bubble is embedded in a piece of glass. For a ray of light passing through the bubble, it behaves like a

(a) converging lens (b) diverging lens
(c) mirror (d) thin plane sheet of glass

SECTION E

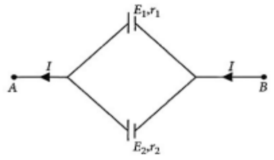
31. (a) Explain the term drift velocity of electrons in a conductor .Hence obtain the expression for the current through a conductor in terms of drift velocity.

(b) Two cells of emfs E_1 and E_2 and internal resistances $r_1$and r_2 respectively are connected in parallel as shown in the figure. Deduce the expression for the

(i) equivalent emf of the combination

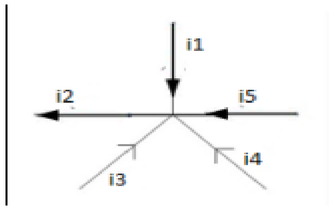
(ii) equivalent internal resistance of the combination

(iii) potential difference between the points A and B.

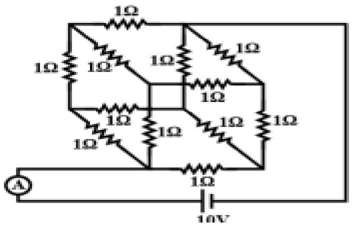


Or

a) What is the relation between currents in the figure below. Define the law used in establishing the relation.



b) A battery or 10 V and negligible internal resistance is connected across the diagonally opposite corner of a cubical network consisting of 12 resistors each of resistance 1Ω. The total current 1 in the circuit external to the network is



32. (i) Define a wavefront. How is it different from a ray?

(ii) Using Huygens's construction of secondary wavelets draw a diagram showing the passage of a plane wavefront from a denser to a rarer medium. Using it verify Snell's law.

(iii)In a double slit experiment using light of wavelength 600nm and the angular width of the fringe formed on a distant screen is 0.1°. Find the spacing between the two slits.

(iv) Write two differences between interference pattern and diffraction pattern.

Or

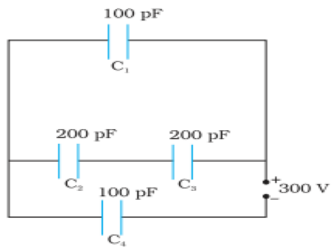
(a) Derive expression for the lens maker's formula using necessary ray diagrams. Also state the assumptions in deriving the above relation and the sign conventions used.

(b) A beam of light converges at a point P. Now a lens is placed in the path of the convergent beam 12 cm from P. At what point does the beam converge if the lens is

(i) a convex lens of focal length 20 cm, (ii) a concave lens of focal length 16 cm ?

33. (i) Derive an expression for the capacitance of a parallel plate capacitor with air present between the two plates.

(ii)Obtain the equivalent capacitance of the network shown in figure. For a 300 V supply, determine the charge on each capacitor.



OR

(i) A dielectric slab of thickness 't' is kept between the plates of a parallel plate capacitor with plate separation 'd' ($t < d$). Derive the expression for the capacitance of the capacitor. (ii)A capacitor of capacity C_1 is charged to the potential of V_0. On disconnecting with the battery, it is connected with an uncharged capacitor of capacity C_2 as shown in the adjoining figure. Find the ratio of energies before and after the connection of switch S.